

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Autonomous Systems [CSE_4023]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain localization technique (2)
A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
B) Identify the reasons for GPS and camera map matching improving the localization (4)
Course Outcome: "CO1" / Cognitive Level: "Application"
C) Examine why localization is more difficult in tunnels (4)
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Explain backscatter (2)
A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
B) List the methodology of reducing backscatter noise in turbid water images? (4)
Course Outcome: "CO2" / Cognitive Level: "Application"
C) Identify reasons with justification why turbidity degrades image clarity? (4)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Explain optical flow (2)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) List suitable options when optical flow fails and sensor can provide motion estimates (4)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Examine the reasons for optical flow's struggle on uniform surfaces (4)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Explain is path planning? (2)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"

B) Identify the most suitable algorithm plans a safe path around obstacles? (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

C) Identify how UAV path planning consider battery constraints? (4)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

5) Explain lane keeping (2)

A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) List the steps involved in detecting lane boundaries (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

C) Analyze why lane-detection systems fail in heavy snow with proper justifications (4)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4023:Autonomous Systems

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4023:Autonomous Systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	20(40%)
4	Analysis	10	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)